

The ordinary Kreweras complement: an exact finite cycle atlas

HCS-C209 source-derived research package

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Abstract

We freeze a finite dynamical system on the noncrossing partitions of a labelled polygon. The clock is one ordinary Kreweras complement, defined by a permutation formula with an explicit orientation. The type-A order- $2n$ cyclic-sieving theorem supplies every iterate fixed count. We then derive, without fitting, all least-period populations, cycle counts, the finite Artin–Mazur zeta, the reciprocal Koopman determinant, root-of-unity spectral multiplicities, rank duality, and polygon-reflection reversors. The actual order is 1, 2, or $2n$ at $n = 1$, $n = 2$, or $n \geq 3$; this small-dimensional boundary is part of the theorem. Independent set-partition enumeration through $n = 8$ and exact SymPy cyclotomic checks through $n = 12$ audit the ledger. The result is deliberately source-local: Catalan data carry no intrinsic rational-prime clock, so no arithmetic determinant or Hilbert–Pólya claim is made.

1 Object and orientation

Place vertices $[n] = \{0, 1, \dots, n-1\}$ counterclockwise on a convex polygon. A set partition is noncrossing when no two blocks contain a, c and b, d in the cyclic order $a < b < c < d$. For $\pi \in \text{NC}(n)$, let p_π cycle each block in increasing order, and let $c = (0\ 1\ \dots\ n-1)$. We use

$$K(\pi) = \text{cycles}(p_\pi^{-1}c). \quad (1)$$

The notation cycles means the set partition formed by the cycles of a permutation. This convention gives the clockwise rotation $\rho_{-1}(i) = i-1 \pmod{n}$ in the square identity below. Replacing c by c^{-1} conjugates the action by a reflection and changes only this sign.

There are two useful orders. Let

$$L_n = \begin{cases} 1, & n = 1, \\ 2, & n = 2, \\ 2n, & n \geq 3, \end{cases} \quad G_n = \begin{cases} 1, & n = 1, \\ 2n, & n \geq 2. \end{cases} \quad (2)$$

L_n is the actual order of the permutation K . G_n is the abstract cyclic-group order in the order- $2n$ CSP. At $n = 2$ the action has a kernel of order two, so these symbols must not be conflated.

2 Structural identities

Theorem 2.1 (ordinary complement identities). *For every n , K is a bijection of $\text{NC}(n)$ and*

$$K^2 = \rho_{-1}, \quad |K(\pi)| = n + 1 - |\pi|. \quad (3)$$

It reverses refinement. Every reflection $R_j(i) = j - i \pmod{n}$ obeys $R_j^2 = 1$ and $R_j K R_j = K^{-1}$. The actual order is L_n in (2).

Proof sketch and boundary check. The permutation factorization $p_\pi p_{K(\pi)} = c$ is the usual geodesic factorization for a noncrossing partition. Taking inverses and multiplying by c gives the complement involution on the geodesic interval, hence the block count and refinement reversal. Applying the factorization twice relabels all vertices by $i \mapsto i - 1$, giving the first identity in (3). Conjugating the displayed permutations by R_j gives the dihedral relation. For $n \geq 3$, polygon rotation acts faithfully on $\text{NC}(n)$, so K^2 has order n . If n is odd, then K^n is an odd power and reverses rank, sending the discrete partition to the indiscrete partition; hence it is not the identity. If n is even, then $K^n = (K^2)^{n/2} = \rho_{-n/2}$, which is nontrivial by the same faithfulness. Thus $K^n \neq 1$ in either parity, while $K^{2n} = 1$, and the order is $2n$. Directly, $\text{NC}(1)$ is a singleton and K exchanges the two elements of $\text{NC}(2)$. $\square$

The rank $r(\pi) = n - |\pi|$ has Narayana distribution

$$\#\{\pi : |\pi| = b\} = \frac{1}{n} \binom{n}{b} \binom{n}{b-1}, \quad |\text{NC}(n)| = \text{Cat}_n = \frac{1}{n+1} \binom{2n}{n}. \quad (4)$$

The rank involution in Theorem 2.1 is checked row by row in the machine-readable evidence.

Audit convention. The distinction between the actual order L_n and the abstract CSP order G_n is part of the statement, not a post-processing choice; in particular, the $n = 2$ generator is evaluated at fourth roots.

3 The fixed-point theorem

Write $[m]_q = 1 + q + \dots + q^{m-1}$ and

$$\text{Cat}_n(q) = \frac{[2n]_q!}{[n]_q![n+1]_q!} \in \mathbb{Z}[q]. \quad (5)$$

The rotation CSP for $\text{NC}(n)$ is the type-A theorem of Reiner–Stanton–White. The order- $2n$ complement extension is the $m = 1$ type-A Kreweras CSP, recorded by Bessis–Reiner and credited there to direct calculations of White. We use these results with attribution, not as a claim of priority.

Theorem 3.1 (all-iterate fixed ledger). *Let $F_n(d) = |\text{Fix}(K^d)|$. For $n = 1$, $F_1(d) = 1$. For $n \geq 2$, reduce d modulo L_n . If $d = 2r$ is even, then*

$$F_n(2r) = \begin{cases} \text{Cat}_n, & r \equiv 0 \pmod{n}, \\ \binom{2g}{g}, & 0 < r < n, \quad g = \gcd(n, r). \end{cases} \quad (6)$$

For odd d , $F_n(d) = \binom{n}{(n-1)/2}$ exactly when n is odd and $d \equiv n \pmod{2n}$; all other odd rows are zero. Equivalently, for $n \geq 2$,

$$F_n(d) = \text{Cat}_n(e^{2\pi i d / G_n}). \quad (7)$$

At $n = 2$, equation (7) uses fourth roots for the abstract C_4 action even though the image permutation has order two.

Root reduction. The CSP gives (7). For even exponents it is the usual rotation specialization: a rotation with $g = \gcd(n, r)$ fixes $\binom{2g}{g}$ partitions, with the identity giving Cat_n . At an odd root the cyclotomic factors in (5) cancel only for the central half-turn; rank reversal also forces $|\pi| = (n+1)/2$, so this row is absent for even n and has the central binomial value for odd n . These elementary cancellations are implemented with integer polynomial remainders in the SymPy audit. $\square$

For orientation, the first rows are

n	L_n	Cat_n	$(F_n(0), F_n(1), \dots, F_n(L_n - 1))$
1	1	1	(1)
2	2	2	(2,0)
3	6	5	(5,0,2,3,2,0)
4	8	14	(14,0,2,0,6,0,2,0)
5	10	42	(42,0,2,0,2,10,2,0,2,0)
6	12	132	(132,0,2,0,6,0,20,0,6,0,2,0)

4 Periods, zeta, and Koopman spectrum

The rest of the atlas is a finite consequence and needs no additional source assumption. Let μ denote the integer Möbius function.

Theorem 4.1 (cycle and determinant package). *For every $\ell \mid L_n$, define*

$$P_{n,\ell} = \sum_{d \mid \ell} \mu(\ell/d) F_n(d), \quad C_{n,\ell} = P_{n,\ell}/\ell. \quad (8)$$

Then $P_{n,\ell}$ is the number of points of least period ℓ and $C_{n,\ell}$ is the number of cycles. They are nonnegative integers and sum to Cat_n . The finite zeta and Koopman determinant are

$$\zeta_{K,n}(z) = \prod_{\ell \mid L_n} (1 - z^\ell)^{-C_{n,\ell}}, \quad \det(I - zU_n) = \prod_{\ell \mid L_n} (1 - z^\ell)^{C_{n,\ell}} = \zeta_{K,n}(z)^{-1}, \quad (9)$$

where $U_n f = f \circ K$ on $\ell^2(\text{NC}(n))$.

Proof. The fixed-point trace identity is $F_n(d) = \sum_{\ell \mid d} P_{n,\ell}$. Möbius inversion gives (8); dividing by ℓ counts cycles. Each cycle contributes the factor $1 - z^\ell$ to a permutation determinant and its reciprocal to the Artin–Mazur product. This proves (9). $\square$

Corollary 4.2 (spectrum and reversor). *The spectrum of U_n consists of L_n th roots, with*

$$\text{mult}\left(e^{2\pi i k/L_n}\right) = \sum_{\substack{\ell \mid L_n \\ L_n \mid k\ell}} C_{n,\ell}, \quad \text{Tr}(U_n^d) = F_n(d). \quad (10)$$

The permutation matrix is unitary. For every j , the map $J_j f = \overline{f \circ R_j}$ is antiunitary and reverses U_n : $J_j U_n J_j = U_n^{-1}$.

For example, the nonzero cycle lengths for $n = 3, 4, 5, 6$ are respectively $\{2, 3\}$, $\{2, 4, 8\}$, $\{2, 5, 10\}$, and $\{2, 4, 6, 12\}$. The machine ledger stores every exponent rather than only these summaries.

5 Independent evidence

The producer emits exact rows for $1 \leq n \leq 24$ (597 fixed rows, 131 period rows, 597 spectral rows, 300 rank rows, and 24 structural rows). It also emits q-Catalan coefficients through $n = 12$. A separate checker reimplements restricted-growth set partitions, crossing detection, the permutation in (1), all reflections, and the cycle decomposition. It enumerates all 2,055 partitions for $1 \leq n \leq 8$ and reports

8,025 assertions. It does not import producer code. A separate SymPy program reconstructs the q -factorial quotient and cyclotomic remainders (1,110 exact checks), then independently recomputes the period and spectral rows through $n = 24$.

The byte-replay script regenerates identical JSON. The mutation harness tests 32 repaired-hash payload mutations and one stale-hash mutation; all are rejected. The release manifest excludes LaTeX sidecars and itself, and stores the hashes of exactly 27 payload files, including the three round PDFs.

Release invariant. Every displayed finite identity is recomputed from integer rows before the manifest is written; the round PDFs are retained so that later edits cannot silently replace the audited manuscript.

6 Scope, relation to C187, and conclusion

C209 is a complete finite combinatorial atlas, not an arithmetic promotion. The integer n , Catalan/Narayana counts, and polygon roots have no intrinsic rational-prime carrier. We therefore record the conservative evaluator tuple

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),$$

with overall verdict ‘ROUTE_A_REJECTED’ and no Route-B invocation. The only operator is the finite source Koopman permutation; no target divisor, functional equation, Euler factor, root number, automorphy statement, or Hilbert–Pólya operator is asserted.

C187 instead owns promotion of rectangular standard Young tableaux, its q -hook polynomial, and evacuation. C209 owns set partitions, geometric complement, the K^2 polygon rotation, and the order- $2n$ complement clock. Shared Möbius/CSP bookkeeping does not make one a parameter continuation of the other.

References

- [1] G. Kreweras, “Sur les partitions non croisées d’un cycle,” *Discrete Mathematics* 1 (1972), 333–350.
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- [3] D. Bessis and V. Reiner, “Cyclic sieving of noncrossing partitions for complex reflection groups,” *Annals of Combinatorics* 15(2) (2011), 197–222.